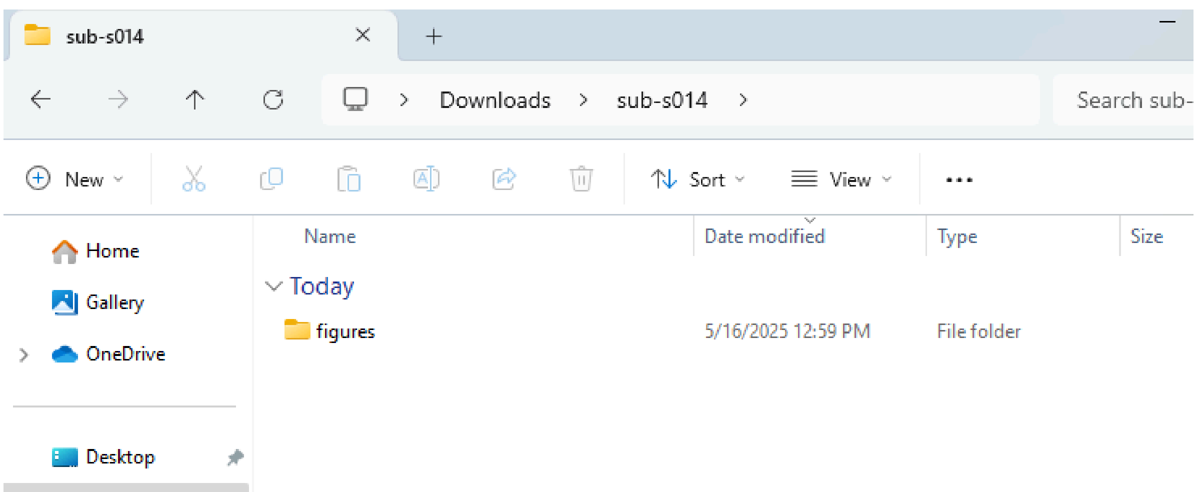
Brains QC (Thank you!!)

1. **Big picture**
    a. Data-quality effects tend to be orders of magnitude larger than psychiatric effects on brain signal.
    b. MRI has a troubled history of [exciting findings later being attributed to] problems with image quality that vary between groups. We have better systems now, but this history often haunts reviewers.
    c. Regions of particular interest for MDD (i.e., subgenual anterior cingulate) can be particularly difficult to image due to disruptions in [clean electromagnetism]
    d. Because all analyses build on the assumption that we are analyzing "true" brain signal, QC can make or break entire projects.
2. **Getting started**
    a. Login to Sherlock On Demand: https://ondemand.sherlock.stanford.edu
        i. Navigate to **/oak/stanford/groups/nolanw/Brains_data_preproc/fmriprep**
        ii. Download the /figures file within /subject
        iii. Download the .html report
        iv. If you place /figures within an (empty) folder in your downloads directory with the subject name, the .html report will be able to generate all figures locally

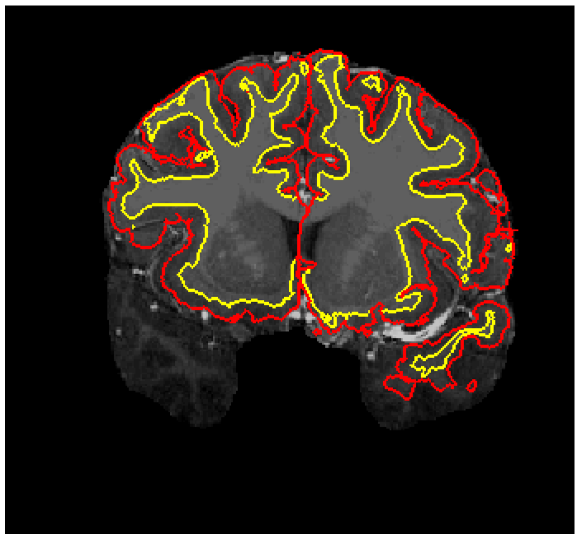


Example "figures" download. With figures in a manually created "sub-s014" folder, sub-s014_anat.html (placed within /Downloads) will be able to locate all image resources needed to evaluate the QC report within.

3. **Structural QC**
    a. Imperfect structural scans are common, but structural artifacts propagate downstream to almost all analyses.
    b. We will rate structural processing on a 1-4 continuum, with 3 or 4s constituting "passes" and 1s or 2s indicating "failure". Within the .html report, we'll look at the image itself, the alignment to a template, and the segmentation of the cortical surface and white matter boundary.

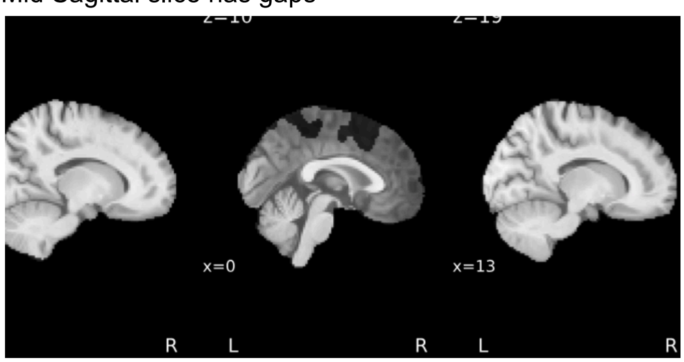
i. 1 = Extremely poor structural image processing. There's no instance in which including data predicated on alignment with this image will allow us to better-assess true brain variability.
ii. 2 = Sub-par image, alignment, or reconstructed surface. Errors are widespread, not one-off instances
iii. 3 = Mostly good structural image, alignment, and surface reconstruction. Deviations may exist but are the exception not the rule
iv. 4 = Pristine structural image, alignment to group-average structural (MNI152), and surface reconstruction

c. Common problems
   i. Cortical surface reconstruction inaccurate



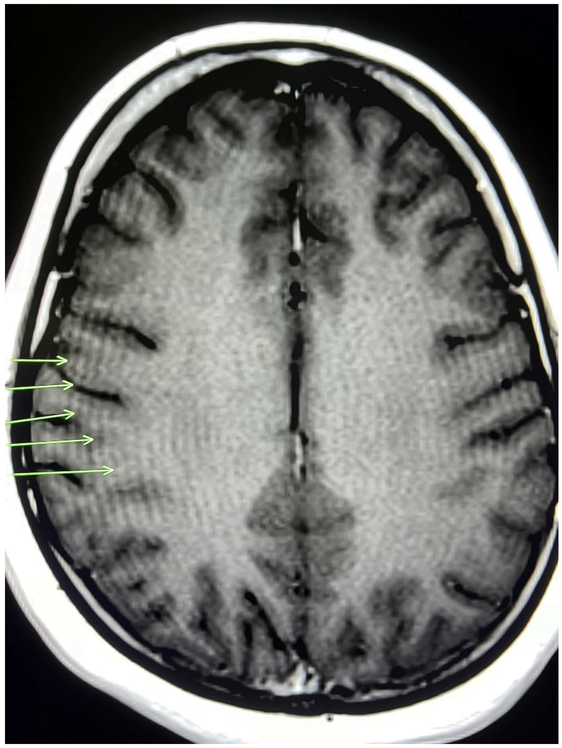
This would be a 1. Although the cortical ribbon is accurate in some places, being grossly misaligned with temporal lobes will hamper registration of every brain area.

   ii. Mid Sagittal slice has gaps



This is actually not a problem. True space in the brain exists here between hemispheres, particularly as individuals get older or if they've had severe psychiatric disorder(s) for a while. The template brain tends to smooth past this variability, but if an individual actually has more space between their hemispheres the only accurate segmentation will omit big chunks like this (or even larger). However, this pattern shouldn't be as common further away from this x=0 slice. An image like this could feasibly score a 4, particularly if the red/blue lines *reflecting cortical ribbon segmentation look good.*

iii. Gibbs ringing



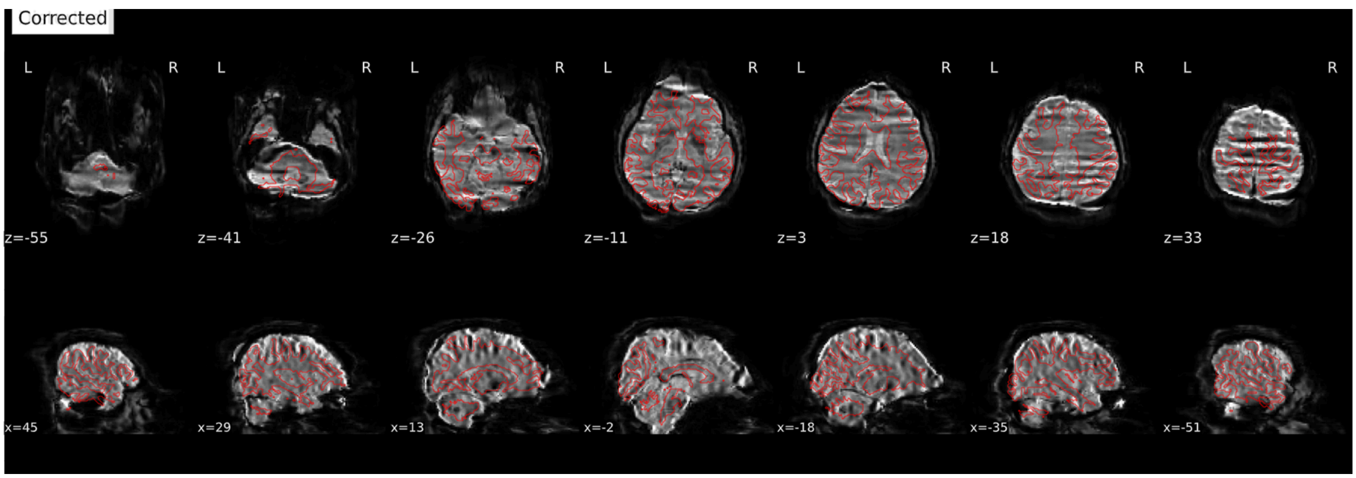
Gibbs ringing can occur near tissue boundaries and comes from fourier transforms over [high *spatial* frequency image] content (i.e., where pixel intensities change quickly over space). As you might have guessed, this is not biologically real (unlike the midsaggital slice *apparent* issue), so a T1 with heavy ringing probably should not get a 4. However, if the surface segmentation looks good, don't hesitate to give it a 3 even if ringing is prominent. This is common enough to not decimate a scan when the artificially hypo/hyper intensities of the rings don't impede gray/white matter boundary detection.

d. Filling out the form
   i. I alphabetically assigned us to "scorer 1-5":
   ii. 1 = Adam, 2 = Andrew, 3 = Lisa, 4 = Niharika, 5 = Tracy
   iii. If there is a 100 for a patient row in your column, that indicates a QC instance randomly assigned to you
   iv. If there is a 0 for a patient row in your column, you do not need to QC that subject's structural
   v. 100's and 0's were assigned so we'd all have 31 or 32, which allows for 3 ratings per subject.
   vi. There is a notes column if you would like to leave notes, but don't feel obligated.
   vii. Note we are *just evaluating structural images now (as of 5/16). SDC correction will have to be redone with Ants SyN, and we will use xcpd outputs for BOLD QC.*

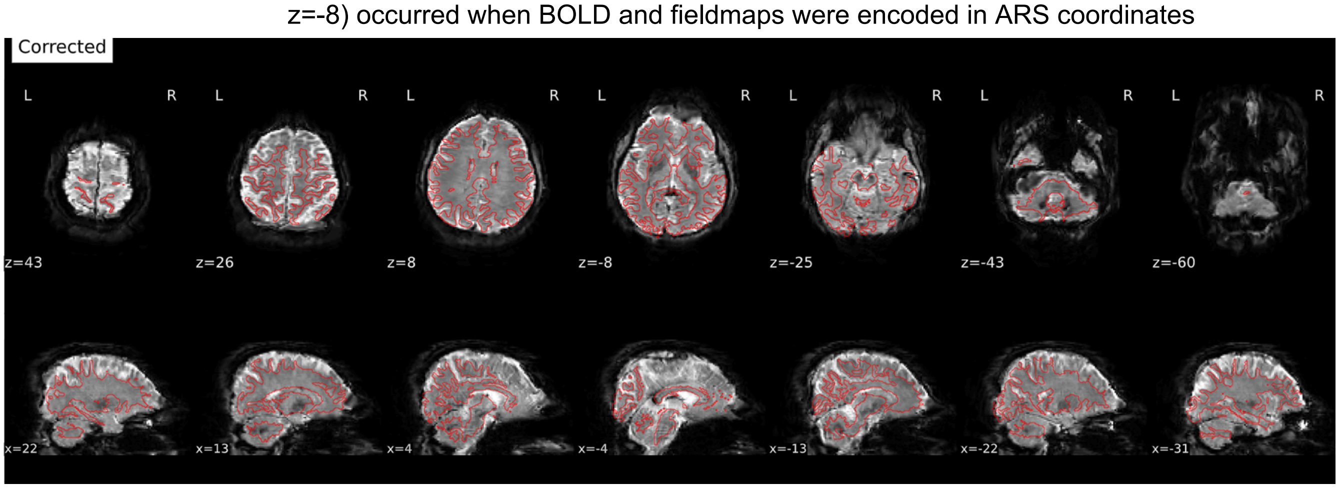
QC spreadsheet:

https://docs.google.com/spreadsheets/d/1p1EJD5CtJ3gtL0ekhPb7oZ4YeuofVBaNplm6qMesf-w/edit?usp=sharing

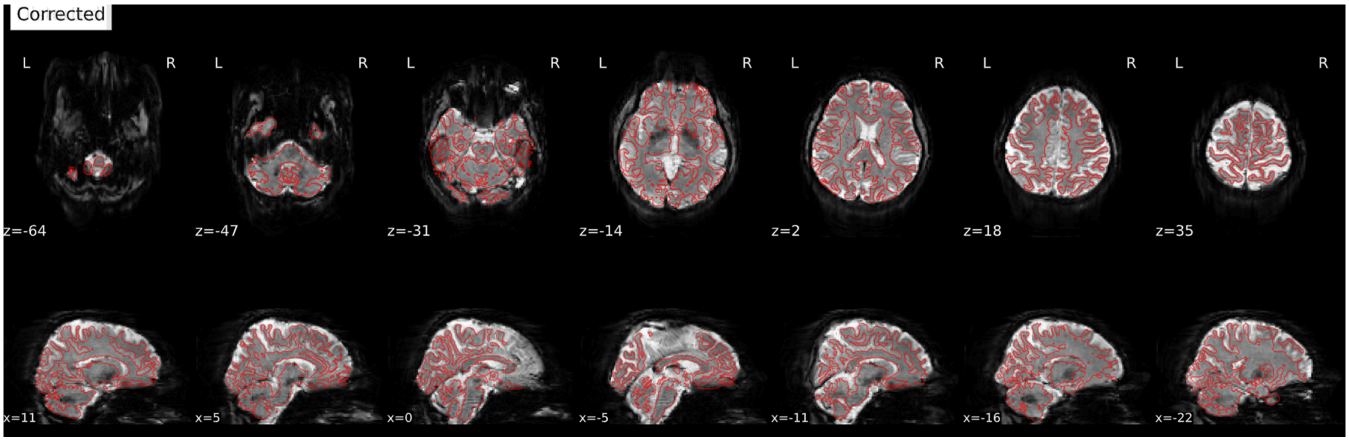
4. **Functional QC**
    a. Functional images can be harder to evaluate, as they are both lower resolution and dynamic over time
    b. Consequently, metrics like framewise displacement (FD; a measure of head motion) or temporal signal-to-noise ratio (tSNR; a measure of signal variation relative to background variation) are often quantified to evaluate functional images.
    c. FD and tSNR can be automatically quantified. We can choose to omit high FD scans and low tSNR scans, or apply “masks”
        i. high motion frames can be censored and low motion frames retained within a scan
        ii. Low tSNR locations can be masked and high tSNR locations retained within a scan, but this is typically done at a study-wide level to have all the same brain locations for each subject
    d. Because there are many publications documenting the efficacy of these automated metrics, <u>our manual QC will only entail catching egregious errors</u>.
    e. Specifically we will look for an issue we have already seen in this dataset: improper susceptibility distortion correction
        i. Issue 1 (catastrophic for images) was occurring when BOLD images were [encoded in ARS coordinates](), but fieldmaps were in RPI



        ii. Issue 2 (bad for images but more subtle, look at boundary of frontal cortex in z=-8) occurred when BOLD and fieldmaps were encoded in ARS coordinates



iii. Susceptibility distortion correction was working when both BOLD and fieldmaps were encoded in RPI. It turns out that all scans were acquired RPI/RPI, but older versions of dcm2iix were failing specifically on GE scanners and sometimes imbuing ARS coordinates. Phase encoding direction was specified in reference to coordinates, causing phase encoding distortion corrections to be inaccurate due to coordinate misspecification. Below is the same scan as ii, but with proper coordinate specification and phase encoding (susceptibility distortion) correction.

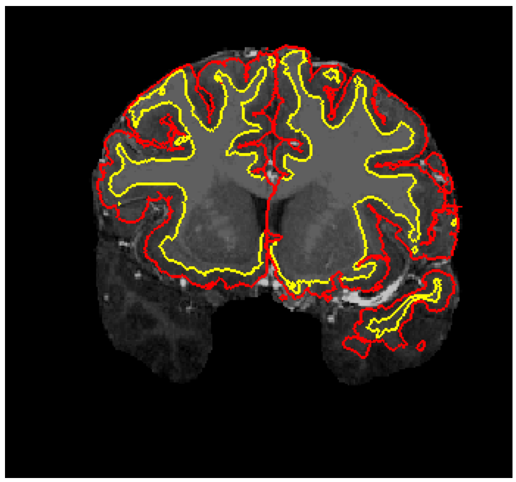


f. Finally and as prior, we will also want to keep an eye out for any *unexpected* issues: problems with images that we have not already noticed in the data. Hopefully nothing comes up here, but please keep your eyes peeled! The most likely scenario is to find something in the file structure itself (i.e., a folder named do not use) rather than a new catastrophic image artifact.

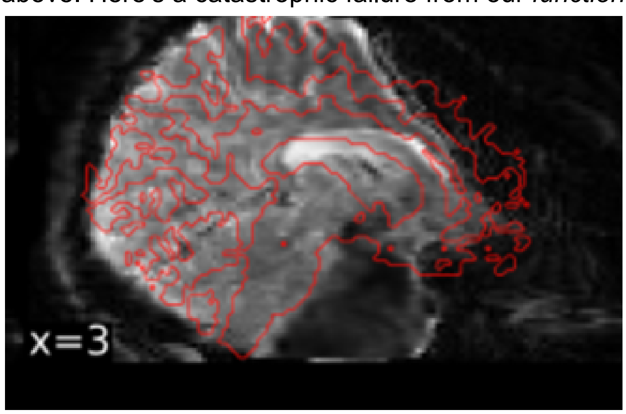
g. Because we will only be looking for obviously flawed images (rather than continuous QC ratings), we'll only need a binary pass/fail and two sets of eyes on each scan. As prior, if a "100" appears in your column then that is a subject you were randomly assigned to.

h. Filling out the form: similar to the last round
    i. I alphabetically assigned us to "scorer 1-4":
    ii. 1 = Adam, 2 = Lisa, 3 = Niharika, 4 = Tracy
    iii. If there is a 100 for a patient row in your column, that indicates a QC instance randomly assigned to you
    iv. If there is a 0 for a patient row in your column, you do not need to QC that subject's scans
    v. 100's and 0's were assigned so we'd all have 24 which allows for 2 ratings per subject.
    vi. We'll use the new .html's i've generated from fMRIprep for QC. These can be found at `/oak/stanford/groups/nolanw/BRAINS_August2025_DataFreeze/fmriprep`
    vii. Please score a subject as "2" for all functional scans pass or "1" for at least 1 functional scan does not pass. If you select 1, please list which scans were messed up (session (V2-V10) and scan number (rs1-rs4)) in the notes column. Notes are only necessary for "fails".
    viii. Feel free to reach out to me (apines@stanford.edu) if you are unsure about any
    ix. Please fill out all of your QC values locally on your PC in a spreadsheet and *then* copy into google sheets upon completion.
    x. THANK YOU AGAIN! Awesome to have you all helping with this.

5. **Individual session structural QC**
   a. We've gone through the averaged subject structural scans already: this covers structural images with respect to their implementation in *functional* preprocessing. Those averaged images are used for functional coregistration, but if we want to quantify structural change over time we need to use the individual structural scans from each session, rather than the structural images averaged over each subject's session
   b. Very similar rules will apply for a good vs. bad image. At a high level, we only want to exclude clearly compromised images. But practically speaking, you two can inclusively flag images that *might* be compromised, and I can come in to verify whether we include or exclude.
      i. **Bad Images**
         1. More specifically, this is the main problem we are looking for (copied from above): the red (gray matter pial surface) and yellow (white matter surface) does not match the structural image.

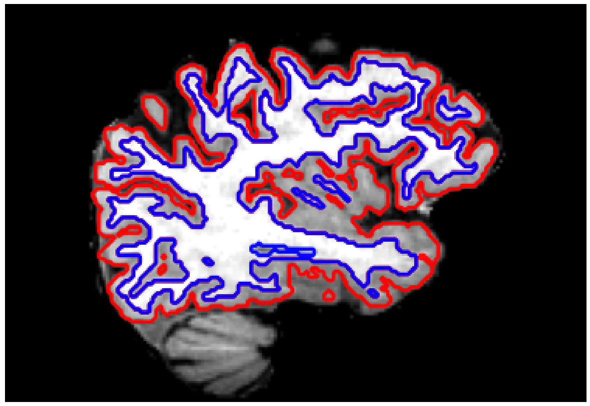


            In this example, you can see poor segmentation in the temporal lobe. Image intensity (brightness) in the structural image is lower in these lobes, which may have led to freesurfer's inability to segment these patches of cortex.
         2. Note we also want to flag catastrophic failures that may not look like the one above! These are quite variable so it's hard to put an example for each one, but if the segmentation or underlying T1-weighted image looks comically bad, please flag it even if it doesn't look like the example above. Here's a catastrophic failure from our *functional* QC for reference:



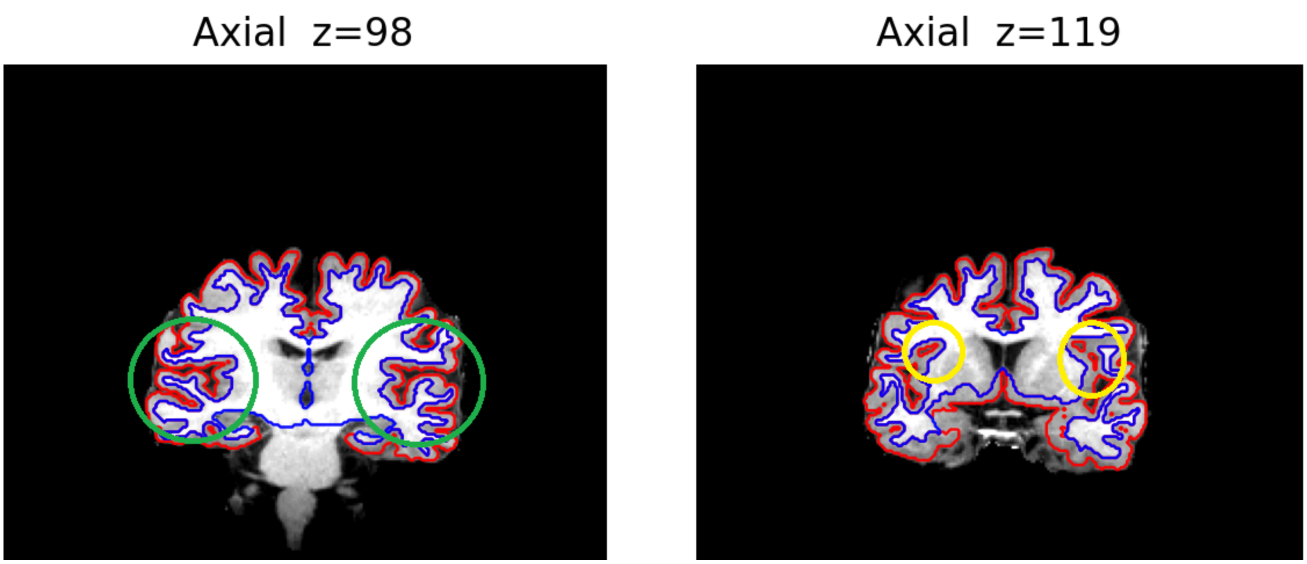
ii. **Good images**

1. This is a boring example of a good segmentation. Gray and white matter boundaries are well-captured, even if you think it could be 3-4% better.



iii. **Good images that may falsely appear to be bad images.** There are three main areas that *are not* cause for alarm that pop up in these types of QC procedures.

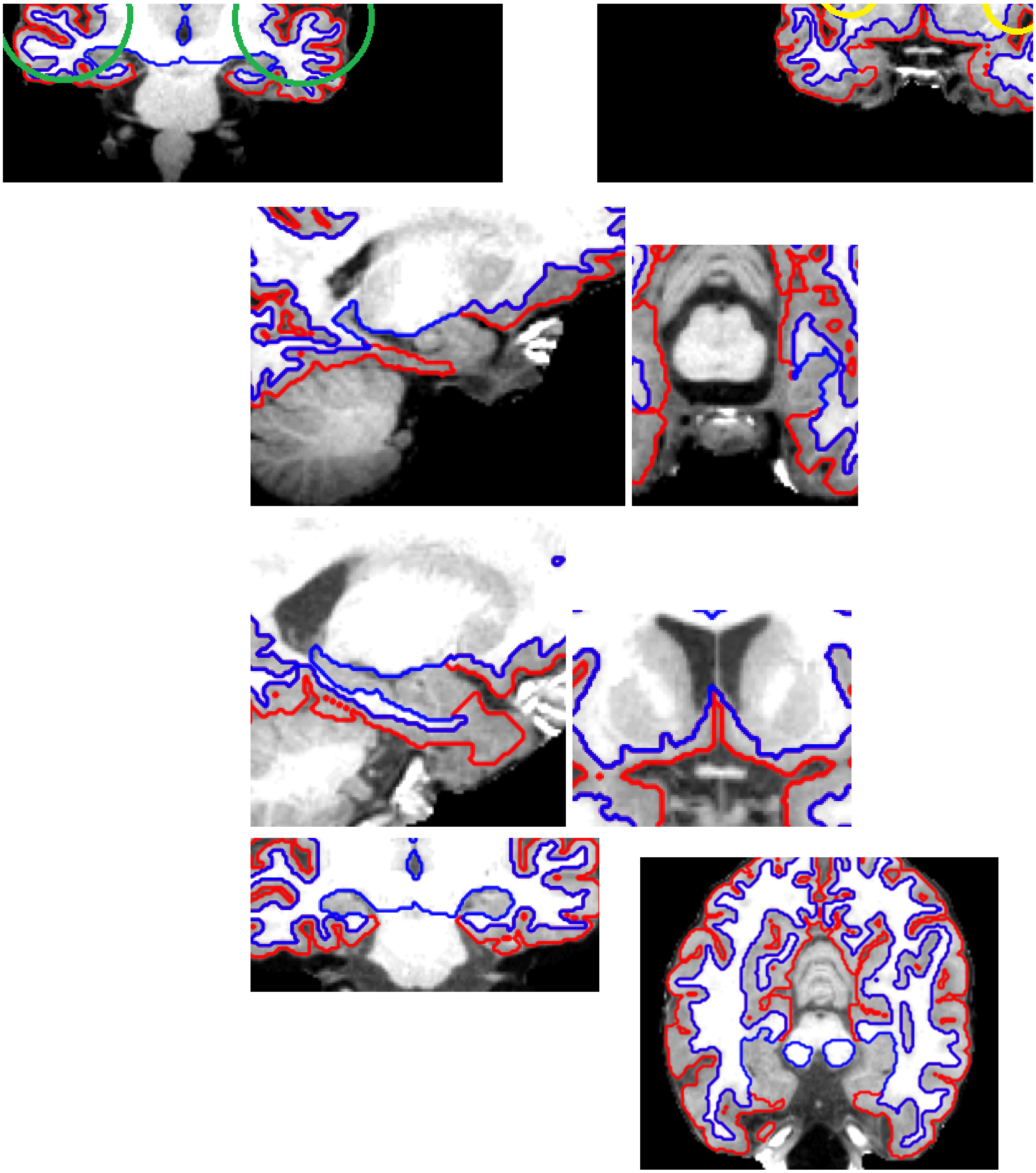
1. The first are the little bubble-like boundaries you can see in the image above. Because we are using slices of images to QC here, it's easy to miss that these small occlusions of non-cortex are usually continuous over many slices, where there is coherent structure. Here's an example:



The isolated circles on the right seem odd and potentially erroneous, but they are actually part of a normal sulcal space that persists across multiple slices in x y and z

2. The second is total failure around basal ganglia, particularly the hippocampus and amygdala. This is normal: these structures are not canonical white or gray matter, so freesurfer processes them separately *after* the step we are visualizing. Here are cortical segmentations that look terrible, but it's because we're not actually trying to segment the hippocampus at this step. So it's *OK* if red lines don't trace out these structures, or the medial striatal structures right above them. All of these

segmentations are OK, because we don't expect them to look right *in this specific part of the brain*

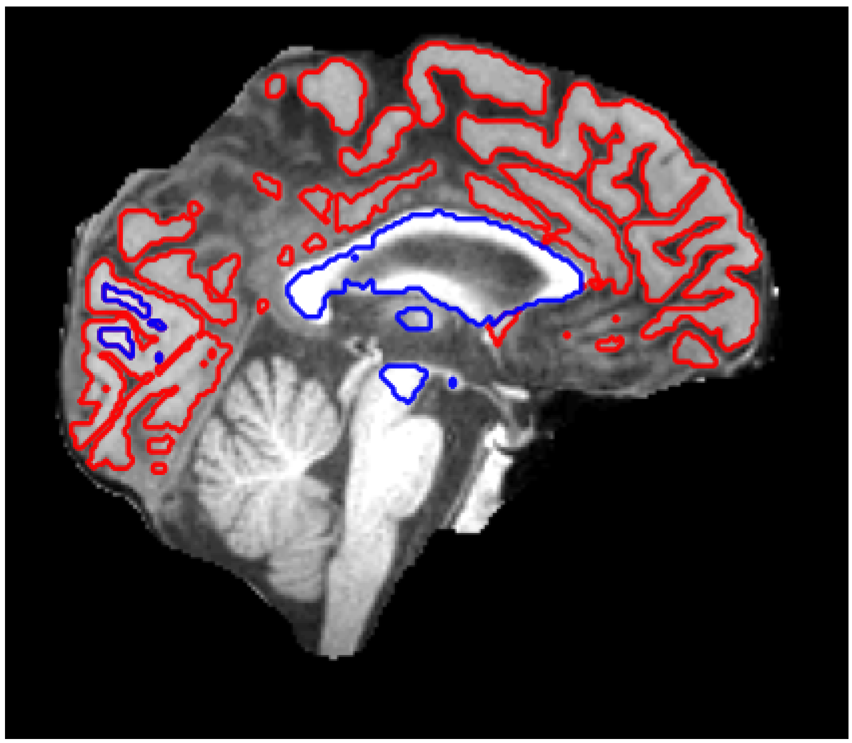


a. Note this is probably the most niche part of QC'ing, because it actually requires a prior on neuroanatomy. These structures are often not super clear to see, so you kind of have to know where

they are. Here are two papers you can check out for a nice hippocampal labeling visualization (figure 1 in both)
https://pubs.rsna.org/doi/full/10.1148/rg.210153
https://insightsimaging.springeropen.com/articles/10.1007/s13244-016-0541-2

Please feel free to reach out if you want a little more reference for what the hippocampus/amygdala/striatum look like in structural images. More than happy to send some more figures/references.

3. The midsagittal slice.This one always looks terrible too, even in good instances. We happen to be capturing a ton of CSF in the slice that falls precisely between cortical hemispheres. Extreme discontinuity of boundaries and exclusion indicated by the ribbons is actually a sign that freesurfer is working well in this longitudinal slice



iv. Downloading the QC materials.
   1. I've tried to streamline this process for this run-through. You should be able to just pull a single .png file per scan from /oak/stanford/groups/nolanw/BRAINS_August2025_DataFreeze/freesurfer/qc_reports which should be adequate for making your ratings. If you want to download a bunch at a time, i'd suggest copying them to your home directory, zipping them into a file, and then copying to your local machine with either sherlock ondemand or scp from the command-line.

v. Filling out the form: entering your QC ratings

1. I’ve added a final tab to our QC spreadsheet: Individual Struct QC
https://docs.google.com/spreadsheets/d/1p1EJD5CtJ3gtL0ekhPb7oZ4YeuofVBaNplm6qMesf-w/edit?usp=sharing
2. Please give each image a “1” if it passes, or a “0” if it does not pass (or if you think it *might* not pass, I can verify).
3. Unlike previous rounds, I’d rather not assign each image to a person. The goal is to have at least 2 of us verify every image. If you flag an image for me to check and I check it, that counts as 2 of us.

   Please just do what you can to the extent that it won't detract from your primary responsibilities. I know we all have other demands on us at the moment. Thanks again and feel free to reach out!